

EDUCATION

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- **University of Georgia – School of Computing** Athens, GA, USA
Doctor of Philosophy in Computer Science - Computer Science;
Aug 2023 - Now
Research Interest: Medical Image processing, Computational Neuroscience, LLMs.
 - **Southeast University – Computer Science and Engineering College** Nanjing, China
Master of Science - Computer Science and Engineering;
Sept 2020 - June 2023
Research Interest: Image processing, Computer Vision, HCI, Neuron reconstruction.
 - **Southeast University – CHIEN-SHIUNG WU College(Honor College)** Nanjing, China
Bachelor of Engineering - Information Science and Engineering; Major GPA: 3.63/4
July 2016 - June 2020
Courses: Signals and System, Data structure, Probability Statistics and Stochastic Process, Digital Signal Processing, Communication Electronic Circuit, Computer Architecture, Statistic Signal Processing, Digital Communication.

PUBLICATIONS

Research Interests: Multimodality Alignment^[1, 2, 3, 7, 8, 9, 10, 11, 12, 13], LLM Agent^[4, 5, 6], Video Generation^[2], Neuron Reconstruction^[3], and Robotics^[10, 12, 14].

1. **Li, Y.**, Liu, Y., Guo, J., Zhao, L., Zhang, Z., Chen, X., ... & Sun, S. (2025). RAU: Reference-based Anatomical Understanding with Vision Language Models. (Under Review), 2025.
2. **Li, Y.**, Kim, S., Wu, Z., Jiang, H., Pan, Y., Jin, P., ... & Li, X. (2024). ECHOPULSE: ECG controlled echocardiograms video generation. ICLR, 2025.
Li, Y., Zhao, H., Jiang, H., Pan, Y., Liu, Z., ... & Liu, T. (2026). Large language models for manufacturing, Journal of Manufacturing Systems, 2026.
Li, Y., Pan, Y., Chen, J., Zhou, Y., Zhao, H., ... & Liu, T. (2026). MEDQUA: A NISQ-Aware Quantum Adapter for Medical Vision Language Models, ISBI(Oral), 2026.
3. **Li, Y.**, Jiang, S., Ding, L., & Liu, L. (2023). NRRS: a re-tracing strategy to refine neuron reconstruction. Bioinformatics Advances, 3(1), vbad054.
4. Dai, H., **Li, Y.**, Liu, Z., Zhao, L., Wu, Z., Song, S., ... & Liu, T. (2023). AD-AutoGPT: An Autonomous GPT for Alzheimer's Disease Infodemiology, PLOS Global Public Health, 2024.
5. Liu, Z., Zhong, A., **Li, Y.**, Yang, L., Ju, C., Wu, Z., ... & Li, X. (2023, October). Tailoring large language models to radiology: A preliminary approach to llm adaptation for a highly specialized domain. In International Workshop on Machine Learning in Medical Imaging (pp. 464-473).
6. Wang, P., Liu, Z., **Li, Y.**, Holmes, J. M., Shu, P., Zhang, L., ... & Liu, W. (2024, July). Fine-Tuning Large Language Models for Radiation Oncology, a Highly Specialized Healthcare Domain. In AAPM 66th Annual Meeting & Exhibition. AAPM.
7. Ma, C., Jiang, H., Chen, W., **Li, Y.**, Wu, Z., Yu, X., Zeng, F., ... & Li, X. (2024). Eye-gaze Guided Multi-modal Alignment Framework for Radiology. NeurIPS 2024.
8. Pan, Y., Jiang, H., Chen, J., **Li, Y.**, Zhao, H., Zhou, Y., ... & Liu, T. (2024). Eg-spikeformer: Eye-gaze guided transformer on spiking neural networks for medical image analysis, ISBI, 2025.
9. Yan, Z., Song, S., Song, D., **Li, Y.**, Zhou, R., Sun, W., ... & Sun, L. (2025). SAMed-2: Selective Memory Enhanced Medical Segment Anything Model. MICCAI, 2025.
10. **Li, Y.**, Wu, Z., Zhao, H., Yang, T., Liu, Z., Shu, P., ... & Liu, T. (2024). ALDM-Grasping: Diffusion-aided Zero-Shot Sim-to-Real Transfer for Robot Grasping. arXiv preprint arXiv:2403.11459.
11. Xiong, F., Xie, P., Zhao, Z., **Li, Y.**, Zhao, S., Manubens-Gil, L., ... & Peng, H. (2024). DSM: Deep sequential model for complete neuronal morphology representation and feature extraction. Patterns, 5(1).
12. Wang, J., Wu, Z., **Li, Y.**, Jiang, H., Shu, P., Shi, E., ... & Zhang, S. (2024). Large language models for robotics: Opportunities, challenges, and perspectives. arXiv preprint arXiv:2401.04334.
13. Hu, H., Shi, E., Yue, C., Yang, S., Wu, Z., **Li, Y.**, ... & Zhang, S. (2025). Harp: Human-assisted regrouping with permutation invariant critic for multi-agent reinforcement learning. ICRA, 2025.

INTERNSHIP

- **United Imaging Intelligence (USA), Research Intern, May 2025 - Aug 2025.**

Led the design of RAU, an end-to-end, pure VLM $\times$ SAM-2 pipeline for reference-guided anatomical understanding. Explored and advanced VLM spatial-relation reasoning; trained with SFT + GRPO; introduced OT-based global matching for box assignment; and engineered a token-to-memory adapter that preserves spatial cues—enabling unsupervised/label-free anatomical understanding and auto-labeling from text or reference images.

Received outstanding mentorship feedback from Dr. Yikang Liu, who described me as “one of the best interns” and noted the work as “very interesting and meaningful.”

- **MGH, Harvard Medical School, CAMCA Lab, Research Intern, May 2024 - Now(Remote).**

Leveraging data from Massachusetts General Hospital (MGH), I developed the first ECG-driven cardiac ultrasound video generation model, which enables heart condition assessment using only wearable ECG devices. This groundbreaking innovation eliminates the reliance on fixed ultrasound equipment, significantly enhancing convenience and expanding application scenarios, particularly in remote or resource-limited environments where traditional imaging tools are inaccessible. The model employs a VQVAE-based approach to convert ultrasound videos into token sequences and integrates a Masked Autoencoder (MAE) training strategy. This methodology successfully produced a high-quality video generation model, achieving state-of-the-art (SOTA) results in the ECHO video generation domain and demonstrating the transformative potential of ECG-guided video synthesis for non-invasive cardiac diagnostics.

Throughout the internship, my work received high praise from Dr. Xiang Li, an Associate Professor at Harvard Medical School, who recognized its innovative approach and real-world implications. Our work was accepted at ICLR 2025, and we are optimistic about its potential impact in advancing cardiac care technologies.

- **Mayo Clinic, Research Intern, May 2024 - Now.**

Leveraging the computational resources provided by Mayo Clinic, I trained a Multi-Modal Large Language Model (MLLM) specifically designed for 3D medical imaging. This effort was aimed at developing a Medical Agent capable of automatically generating diagnostic reports for patient CT and MRI scans. Through the integration of advanced deep learning techniques and large-scale medical datasets, the model was fine-tuned to interpret complex 3D images and translate them into clinically relevant information. This work not only improved the efficiency and accuracy of diagnostic workflows but also provided the research team with a powerful tool for medical decision support, laying the groundwork for future advancements in AI-driven healthcare solutions.

- **Shenzhen Institute of Advanced Technology, Research Intern, March 2023 - May 2023.**

I integrated my NRRS algorithm and workflow optimizations into the neuron-reconstruction pipeline, resolving key bottlenecks and significantly improving result quality. I profiled failure modes, tuned parameters to the lab's data characteristics, and refactored stages for throughput and stability. The updated pipeline now delivers higher accuracy and reliability with measurable gains in end-to-end efficiency.

- **Huawei, Software Engineer Intern, June 2020 - Sep 2020.**

Work as a telecommunication algorithm intern that uses AI algorithms to reduce the signal-to-noise ratio at the transmitter end of the Massive-MIMO antenna system.

PROJECTS

- **AD-AutoGPT:** I developed the first Alzheimer's research agent leveraging the LangChain framework. By employing techniques such as goal decomposition and constructing an instruction library, the LLM agent can automatically collect, analyze, summarize, and synthesize the latest research advancements in the academic field.
- **LDM-Grasping:** I enhanced the diffusion model by incorporating an additional loss derived from the discriminator of a GAN model. This approach ensures that the generated images maintain the contours and positions of objects consistent with those in the initial condition image. The method enables the diffusion model to transform simulated data into realistic images in a controlled manner, addressing the Reality Gap problem in robotics applications.
- **Deid-GPT:** Medical data involves strict privacy concerns. We fine-tuned the LLaMA 3-70B large language model using a combination of synthetic data and de-identified real data, enabling the model to identify and anonymize personal information in medical reports. Similarly, we fine-tuned a vision-language model (VLM) using the same approach, allowing it to anonymize patient-identifiable information in medical videos effectively.

HONORS AND AWARDS

- Awarded 2500+ citations on Google Scholar.
- The Second Prize of the National Mathematical Modeling Competition in 2018 (1st pick in SEU).
- Champion of the IEEE standard MicroMouse Maze Competition at SEU in 2018.

SKILLS AND HOBBIES

- **Programming language :** C/C++, Python, JAVA, JS, matlab, NEURON.
- **Sports:** Leader of college basketball team, Badminton, tennis.
- **Musical instrument:** Ukulele, Violin.